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Editor-in-Chief

Sports Medicine

Dear Mr. Steve McMillan,

We are pleased to submit our manuscript titled “*Prevalence, Reporting Practices, and Methodological Quality of A Priori Power Analyses in Sports and Exercise Science Research*” for consideration in *Sports Medicine*.

Conducting an a priori power analysis is essential for designing studies with adequate statistical power, thereby increasing the informativeness and replicability of research findings. In contrast, low-powered studies—particularly when combined with selection bias—can yield inflated effect sizes and false-positive findings, contributing to the low replication rates observed in initiatives such as the Sports Science Replication Project.

Prior work in sports and exercise science has highlighted a critical concern: the average statistical power of hypothesis-testing studies has been estimated to be approximately 11%. One potential explanation for this low level of statistical power is the limited use of a priori power analyses. Despite their importance, only 41% of hypothesis-testing studies reported conducting an a priori power analysis, and many of these analyses were incomplete or computationally non-reproducible due to missing or insufficiently reported information. Moreover, reporting a power analysis was not associated with larger sample sizes or a higher proportion of supported hypotheses, suggesting that many reported analyses may have been poorly conducted and did not meaningfully inform study design.

We believe our work will be of considerable interest to your readership, as it contributes to ongoing efforts to strengthen the reproducibility and reliability of the evidence base. We confirm that this manuscript is original, has not been published elsewhere, and is not under consideration by any other journal. All authors have approved the submission, and we have disclosed any potential conflicts of interest.

We appreciate your time and consideration and look forward to the possibility of contributing to *Sports Medicine*.

Sincerely,

Cristian Mesquida